

CareLoop

The companion that **actually checks in.**

An agentic AI that phones older people about their medications, listens to how they answer, and decides on its own when their family needs to know.



Reminders can't hear the one word that matters.

WHAT A REMINDER APP RECORDS

“Yes, taken.”

A tap on a notification. No way to tell a confident yes from a guess, or a pill taken from a pill remembered.

WHAT SHE ACTUALLY SAID

“I **think** I took it.”

On a blood thinner, that uncertainty is the signal. Twice in a week, it's a pattern someone should know about.

Reminders only ping

They tell someone to do a thing. They never find out whether it happened.

Families find out late

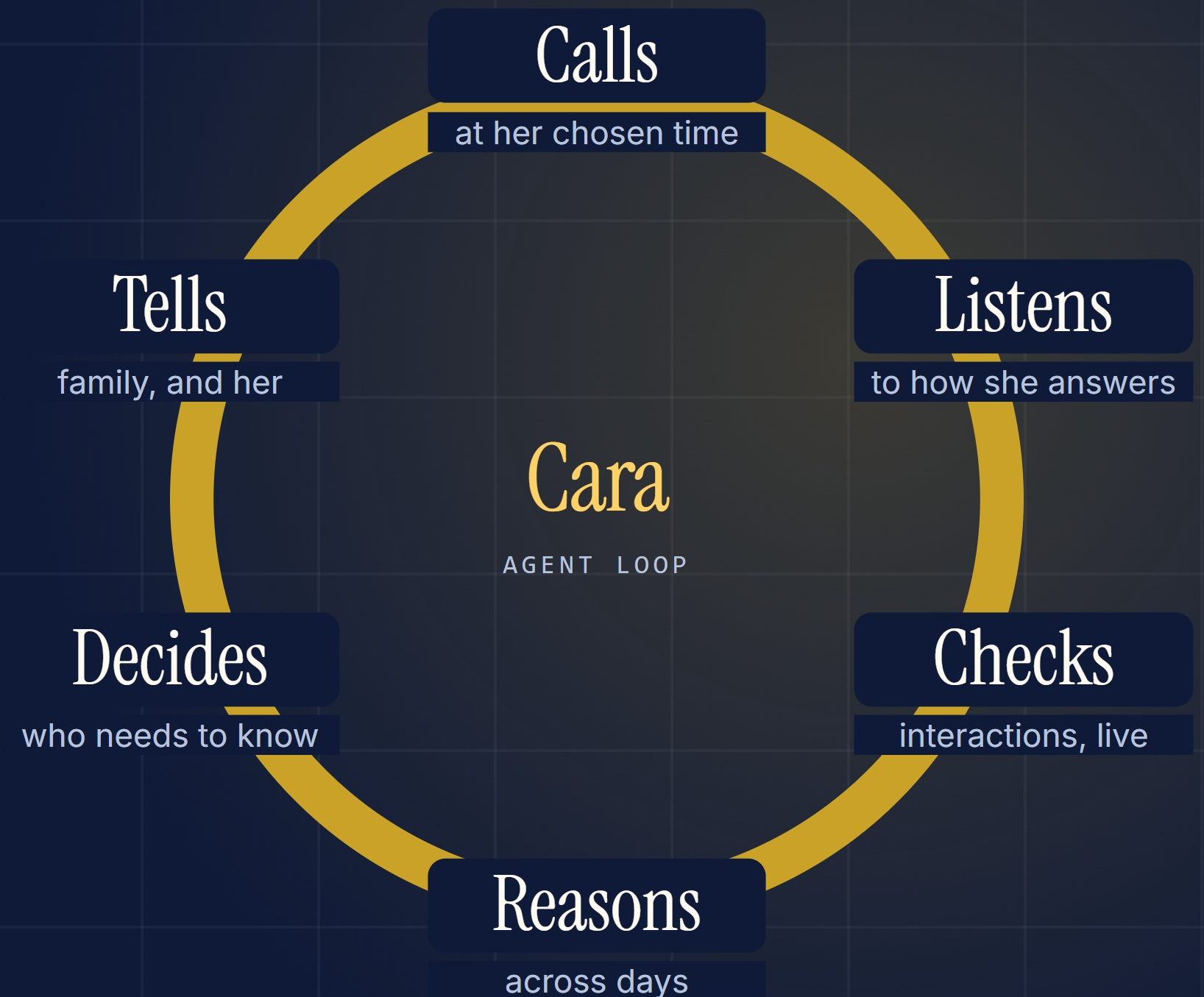
A missed pattern usually surfaces at the pharmacy, or the hospital.

Monitoring feels like surveillance

Nearly half of older adults are uncomfortable being monitored, even when the safety benefit is clear.

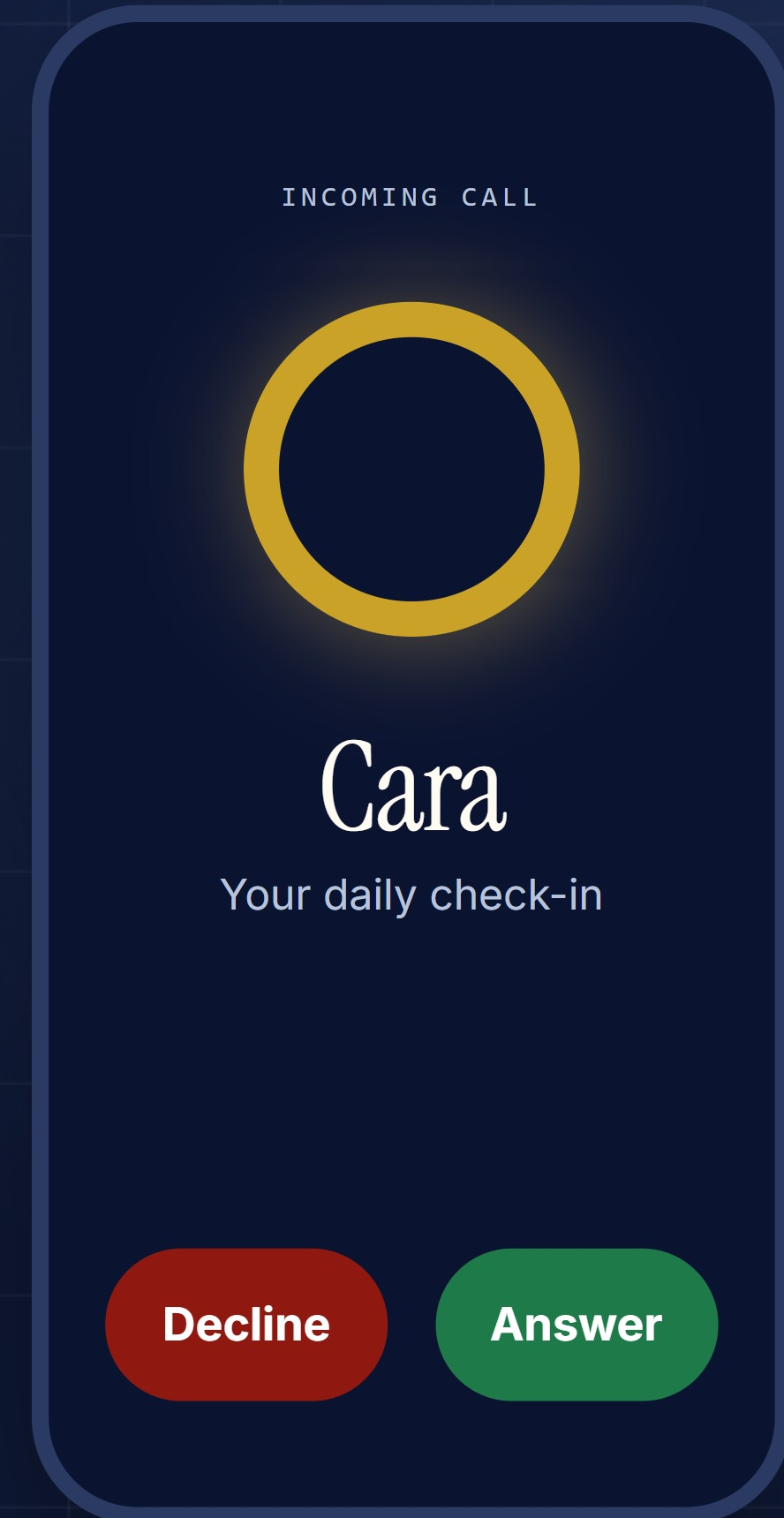
Not a reminder. **A** **phone call** that thinks.

Once a day, at a time she chose, her phone rings. Cara, the agent, runs a full loop on every call, and the loop is the product.



It rings like **a real call**.

- 1 The server rings the phone**
A scheduler sends a high-priority push. Android shows a full-screen incoming call, on the lock screen.
- 2 A live voice conversation**
Gemini Live, streamed directly from the phone with a single-use token that expires in 15 minutes. The API key never ships in the app.
- 3 Cara acts through tools, mid-sentence**
Check an interaction, record a dose, log a reading, note something for next time, report an emergency.
- 4 The call is written, then reasoned about**
The check-in is saved and the reasoning engine decides what happens next.



It checks while she's still talking.

LAYER 1

Curated rules

Deterministic and offline. Never missed on a timeout.

LAYER 2

openFDA label text

Searched against her real medication list.

LAYER 3

RxNorm matching

For drug names said imprecisely out loud.

Declared non-blocking, so Cara never goes silent while a lookup runs. A silent gap on a phone call sounds like the line dropped.

INTERACTION FOUND

Warfarin + ibuprofen

Together they raise the risk of bleeding. Cara explains why, and suggests asking her pharmacist. She never changes a dose.

CHECKED MID-SENTENCE

My knee's been bad, so I've taken ibuprofen.

Thank you for telling me. Let me look at that alongside your other tablets while we talk.

A concern score, not a counter.

CRITICALITY

1.0 → 0.15

A missed blood thinner weighs almost seven times a routine tablet.

REPETITION

n^{1.35}

Two misses is not twice one. It's where "forgot" stops being the best explanation.

UNCERTAINTY

+0.7

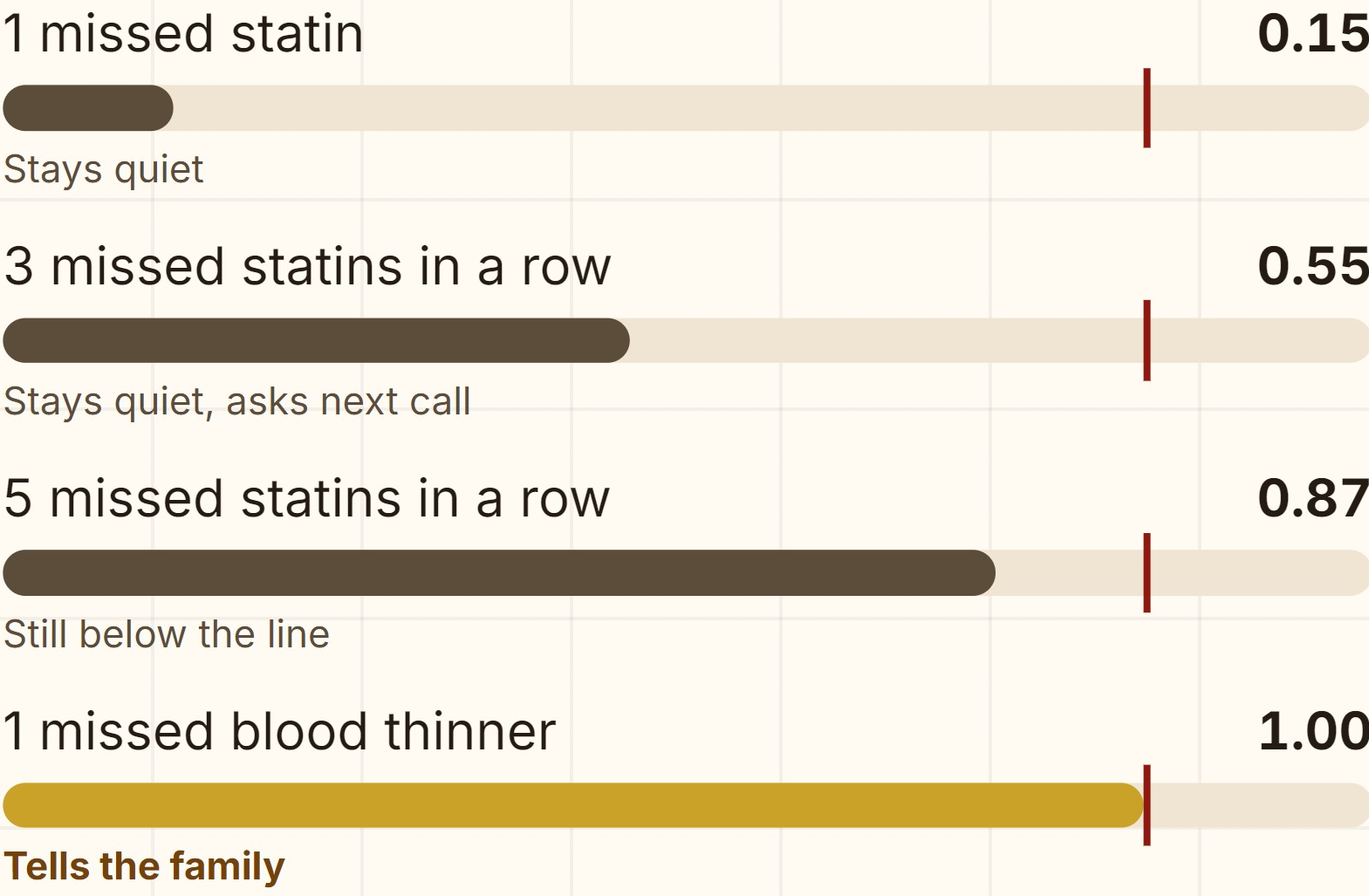
"I think I took it" scores higher than a clean miss.

RECENCY

7 days

Evidence fades by whole days, so a bad week last month stops alerting.

RED LINE • TELL THE FAMILY AT 1.00



Real outputs of the engine, from its own evaluation harness.

Every decision shows **its working.**

CARA GOT IN TOUCH • NEEDS ATTENTION

Margaret has missed her warfarin twice this week.

- Warfarin missed Tuesday and Thursday
Cites both check-ins
- Unsure both times, not a clean miss
Her words: "I'm not sure if I took it or not"
- A blood thinner, where waiting carries real risk
Criticality: critical

Confidence **quite sure** Considered **waiting one more day**

At most three reasons

Each tied to a specific check-in you can open. A trace you can't click through to the evidence is not a trace.

Confidence in words

Never a percentage. And the alternatives it weighed are shown too.

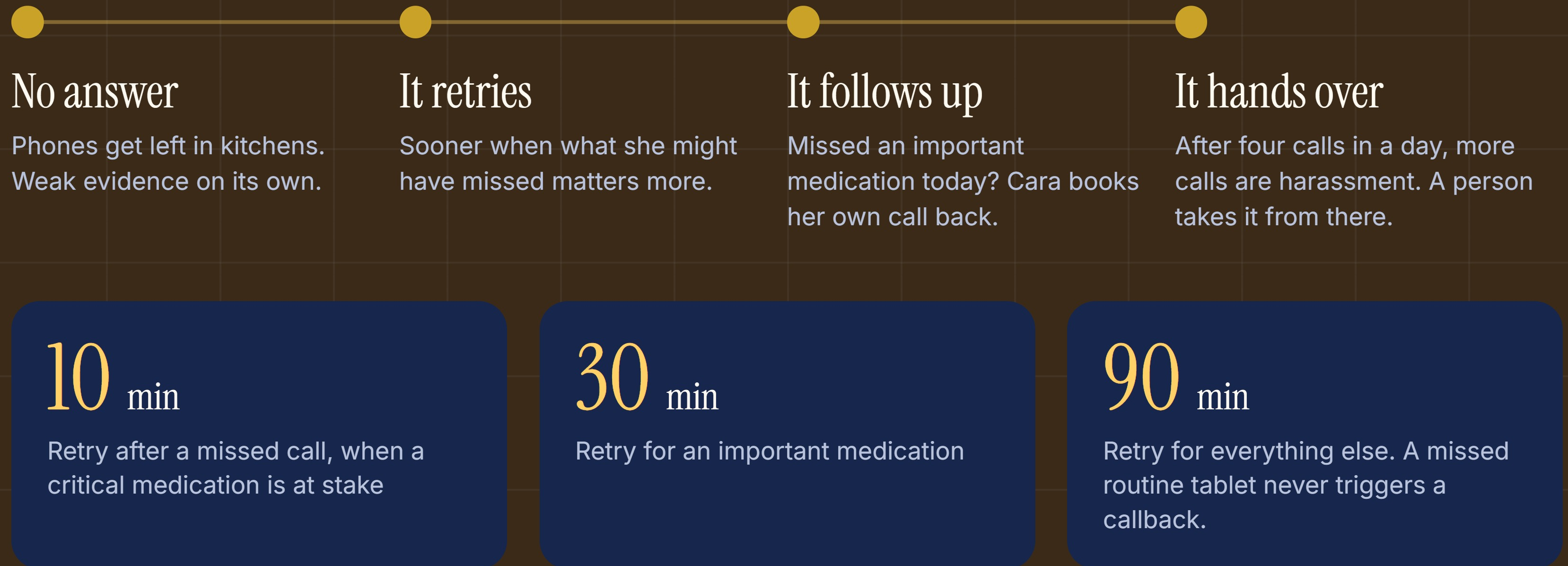
Templated, not generated

The explanation states exactly what the engine weighed. A fluent paraphrase that misstates the reasoning is the failure this product exists to avoid.

Restraint is recorded too

Deciding not to tell anyone is written down, with the score and why.

It knows when to chase, and when to stop.



Never behind her back.

SARAH'S DASHBOARD

Margaret has missed her warfarin twice this week.

She confirmed this in her own app.

MARGARET'S PHONE, SAME MOMENT

I told Sarah you'd missed your warfarin on Tuesday and Thursday.

That's right

I'd add something

Same words, same moment

The family's alert and her copy are one database write. One cannot exist without the other.

Her side of the story

She can agree, or add her account, and it appears beside Cara's.

Her choice what's shared

Mute routine categories. A genuine emergency always reaches family, and she knows that.

Access she can take back

She removes anyone, any time, and sees what Cara remembers about her.

It remembers her, not just her pill box.

HER OWN THREADS

What Cara said she'd come back to

Cara opens and closes her own follow-ups, and chooses when to raise each one again.

Knee pain after starting ibuprofen
• follow up in 2 days

CONVERSATION MEMORY

What the call was actually about

Each call is summarised into what she talked about, written so she'd be comfortable reading it.

Sleeping badly this week.
Daughter visiting on Sunday.

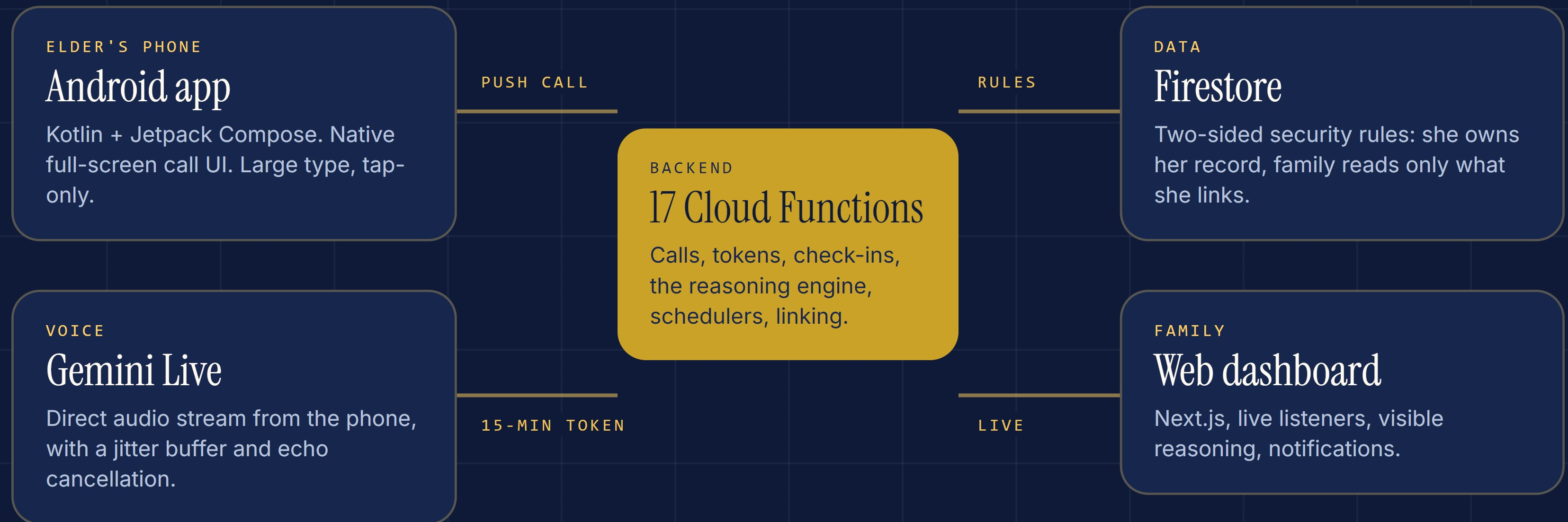
GETTING TO KNOW HER

Conditions, learned by asking

She's asked once, gently, what she keeps an eye on, then Cara asks the right things: blood pressure, sugar, breathing.

"Last time you said your knee was playing up. How is it today?"

Built for a **real call**, not a demo.



Verified on real accounts, against the live backend.

1.3s

Scheduler to ringing phone

The daily scheduler fired and the phone rang 1.3 seconds later.

1.7 → 3.94

Autonomous escalation

One missed blood thinner escalated on its own. A second escalated harder, to urgent.

4 steps

The call loop closes

deliver → token → check-in → outcome, every call.

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Security-rules tests

Run on CI against the Firestore emulator, including the exact query the dashboard runs.

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Reasoning invariants

Never escalates on a routine miss, always cites evidence, same evidence gives the same decision.

Real phone

She heard him

A live call on a physical handset: speech understood, missed dose recorded, memory thread opened by the agent.

Hard lines *it never crosses.*

- ✓ **Never diagnoses, never changes a dose**
It surfaces known interactions and routes to a pharmacist or GP.
- ✓ **Never invents what it doesn't know**
With no medications on record, it asks what she takes instead of guessing.
- ✓ **Private by construction**
No names, medications or transcripts ever reach the logs. The API key never leaves the server.
- ✓ **Emergencies stop the check-in**
Chest pain or a fall: tell her to call emergency services, and alert family at once.
- ✓ **Honest about being an AI**
If asked, Cara says so plainly. Warm, never "sweetie".
- ✓ **Example data is always labelled**
A health record that could be fake must say so. Real data is the default.

From working prototype to real families.

TODAY

Live and working

Android app, live calls, reasoning, escalation and the family dashboard, all deployed.

NEXT

Alerts anywhere

Push notifications that reach family with the browser closed.

THEN

A pilot

A small group of families, measuring whether patterns are caught earlier.

LATER

Care teams

A shareable summary for pharmacists and GPs, with her consent.



Someone checks on your mother **every morning.**

careloop--careloop-adam.europe-west4.hosted.app

BUILT BY ADAM AHMED • DDS AGENTIC AI CHALLENGE